Exercise: Data Visualization – California Housing Dataset Analysis

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1. Task 1: Feature Distribution Visualization with Kernel Density Estimation

1.1 KDE of features

1.2 Method And parameters to set Kernel width

1.3 Discussion Of Distribution

1.4 Interesting Phenomenon

For each continuous feature, present a KDE plot and summarize key observations:

- `median_house_value`: Describe skewness (e.g., right-skewed, capped at $500,000), peaks, and any truncation artifacts.
- `median_income`: Describe distribution shape (e. g. , right- skewed, concentrated in lower-middle income brackets).
- `housing_median_age`: Describe modality (e.g., roughly uniform or bimodal) and range.
- `total_rooms` / `total_bedrooms`: Describe extreme right skewness (many small values, few large values).
- `population` / `households`: Describe right skewness and heavy tails.
- `latitude` / `longitude`: Describe geographic clustering (e.g., peaks corresponding to major urban areas like Los Angeles, San Francisco).
- Highlight interesting patterns (e. g. , geographic clustering of high house values, extreme skewness in room/population features).

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3. Task 2: Bimodal Quantitative Features

3.1 Identification of Bimodal Features

- List quantitative features with clear bimodal KDE distributions (e.g., `latitude`, `longitude`, potentially `median_house_value`).
- For each bimodal feature:
  - Show the KDE plot with two distinct peaks.
  - Rationalize the peaks using domain knowledge:
    - **`latitude`** / **`longitude`**: Peaks correspond to major population centers (e. g. , Northern California: San Francisco Bay Area; Southern California: Los Angeles/Orange County/San Diego).

  - **`median_house_value`**: Peaks may reflect affordable inland areas vs. expensive coastal/urban areas, or income stratification.
  - Other features (if bimodal): Link peaks to demographic/geographic factors (e. g., older vs. newer housing stock for `housing_median_age`).

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4. Task 3: Features Where the Mean Is Not Representative

4.1 List of Non-Representative Mean Features

- List features where the arithmetic mean poorly reflects the central tendency (e. g. , `total_rooms`, `total_bedrooms`, `population`, `households`, `median_house_value`).

4.2 Explanation for Each Feature

For each feature:

- **`total_rooms`** / **`total_bedrooms`**: Extreme right skewness (most blocks have few rooms, a small number of blocks have very large numbers of rooms). The mean is pulled upward by outliers, so the median is a better measure of central tendency.
- **`population`** / **`households`**: Similar right skewness; the mean is inflated by highly populated blocks (e.g., urban centers), while most blocks have smaller populations.
- **`median_house_value`**: Right skewness plus a hard upper limit ($\$500,000$ cap in the dataset). The mean is distorted by high-value coastal properties and the truncation, so the median better represents typical house values.
- (Optional) Other features: Explain skewness/outlier effects on the mean.

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5. Task 4: Most Predictive Features for House Value

5.1 Selection of Top 3 Predictive Features

- Propose three features (e.g., `median_income`, `latitude`, `longitude`, or `median_income`, `latitude`, `housing_median_age`).

5.2 Justification

For each selected feature:

- **`median_income`**: Strong direct correlation with house value (higher income areas can afford more expensive housing; this is the strongest predictor in the dataset).
- **`latitude`** / **`longitude`**: Geographic location captures coastal/ urban premium (e. g., proximity to job centers, amenities, natural scenery), which drives large differences in house values across California.
- **`housing_median_age`** / **`total_rooms`**: Reflects housing quality/ desirability (newer homes or larger homes may command higher prices; alternatively, proximity to amenities correlates with age).
- Explain why these three features are expected to outperform others (e.g., `population` is less predictive because it correlates with household size but not directly with housing value per unit).

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6. Conclusion

- Summarize key findings:
    - Most continuous features are right-skewed, making the median a more robust central tendency measure than the mean.
    - Geographic features (`latitude`, `longitude`) are bimodal due to California's dual population centers (Northern/Southern California).
    - `median_income` and location are the strongest predictors of house value.
- Reflect on the utility of KDE for understanding distribution shape and identifying patterns in the dataset.